

Homework 6, Question 4b Explained

Conditional Distribution of Poisson Process

1 Problem Statement

Let $\{N_t\}$ be a Poisson process with rate λ .

Question 4b: For any $t, s \geq 0$, what is the conditional distribution of N_t given $N_s = n$?

Key Observation

This question is asking for the **general formula** for ALL values of t and s , not just one specific case.

The answer depends on the relationship between t and s :

- **Case 1:** $t < s$ (we're asking about an EARLIER time)
- **Case 2:** $t = s$ (same time)
- **Case 3:** $t > s$ (we're asking about a LATER time)

2 Case 1: When $t < s$ (Earlier Time)

2.1 Setup

You're at time s and you know $N_s = n$ arrivals have occurred.

Question: What's the distribution of N_t (arrivals by an *earlier* time t)?

Time: 0 t s
 |-----|-----|

Events: [N_t] [$N_s - N_t$]
 [$N_s = n$]

2.2 Intuition

Think of it this way:

- You have n arrivals total by time s
- Each arrival is “equally likely” to have occurred anywhere in $[0, s]$
- Time t represents $\frac{t}{s}$ of the total time interval

- So each of the n arrivals has probability $\frac{t}{s}$ of being in $[0, t]$

This is like flipping n coins, each with success probability $\frac{t}{s}$!

2.3 Answer for Case 1

Case 1: $t < s$

$$(N_t \mid N_s = n) \sim \text{Binomial}\left(n, \frac{t}{s}\right)$$

Explicitly:

$$P(N_t = k \mid N_s = n) = \binom{n}{k} \left(\frac{t}{s}\right)^k \left(1 - \frac{t}{s}\right)^{n-k} \quad \text{for } k = 0, 1, \dots, n$$

Note: Question 4a was exactly this case with $t = 4$, $s = 9$, $n = 13$, $k = 5$!

2.4 Properties for Case 1

- **Mean:** $\mathbb{E}(N_t \mid N_s = n) = n \cdot \frac{t}{s}$
- **Variance:** $\text{Var}(N_t \mid N_s = n) = n \cdot \frac{t}{s} \cdot \left(1 - \frac{t}{s}\right) = n \cdot \frac{t}{s} \cdot \frac{s-t}{s}$
- **Support:** N_t can be $0, 1, 2, \dots, n$ (can't exceed n !)

3 Case 2: When $t = s$ (Same Time)

3.1 Setup

You're asking: "Given $N_s = n$, what is N_t when $t = s$?"

3.2 Answer for Case 2

Case 2: $t = s$

$$(N_t \mid N_s = n) = n \quad \text{with probability } 1$$

This is a **degenerate distribution** (deterministic).

Why? If $t = s$, then $N_t = N_s = n$ for certain. There's no randomness!

You can think of this as $\text{Binomial}(n, 1)$ where every "trial" succeeds, so you always get n .

4 Case 3: When $t > s$ (Later Time)

4.1 Setup

You're at time s and you know $N_s = n$ arrivals have occurred.

Question: What's the distribution of N_t (arrivals by a *later* time t)?

Time: 0 s t
 |-----|-----|
 Events: [$N_s=n$] [$N_t - N_s$]
 [N_t]

4.2 Key Insight

We can write:

$$N_t = N_s + (N_t - N_s)$$

We **know** $N_s = n$ (given), and $(N_t - N_s)$ represents **future arrivals** in $(s, t]$.
 By the **memoryless property** of Poisson processes:

- Future increments are **independent** of past history
- So $(N_t - N_s)$ is independent of $N_s = n$
- $(N_t - N_s) \sim \text{Poisson}(\lambda(t - s))$ regardless of what happened before time s

4.3 Answer for Case 3

Case 3: $t > s$

$$(N_t \mid N_s = n) = n + \text{Poisson}(\lambda(t - s))$$

More explicitly:

$$(N_t \mid N_s = n) \sim n + Y \quad \text{where } Y \sim \text{Poisson}(\lambda(t - s))$$

Or writing out the PMF:

$$P(N_t = k \mid N_s = n) = \begin{cases} \frac{e^{-\lambda(t-s)} [\lambda(t-s)]^{k-n}}{(k-n)!} & \text{if } k \geq n \\ 0 & \text{if } k < n \end{cases}$$

Important: N_t cannot be less than n because we already have n arrivals by time s !

4.4 Properties for Case 3

- **Mean:** $\mathbb{E}(N_t \mid N_s = n) = n + \lambda(t - s)$
- **Variance:** $\text{Var}(N_t \mid N_s = n) = \lambda(t - s)$
- **Support:** N_t can be $n, n + 1, n + 2, \dots$ (must be at least n !)

4.5 Why This Makes Sense

Interpretation:

“I know there were n arrivals by time s . How many will there be by the later time t ?”

Answer:

- We **definitely** have the n arrivals that already happened
- **Plus** however many more arrive in $(s, t]$
- Future arrivals follow $\text{Poisson}(\lambda(t - s))$ because the process “restarts” at time s

5 Complete Answer to 4b

Complete Answer to Question 4b

For a Poisson process $\{N_t\}$ with rate λ , the conditional distribution of N_t given $N_s = n$ is:

Case 1: If $t < s$

$$(N_t \mid N_s = n) \sim \text{Binomial}\left(n, \frac{t}{s}\right)$$

Case 2: If $t = s$

$$N_t = n \quad (\text{deterministic})$$

Case 3: If $t > s$

$$(N_t \mid N_s = n) = n + \text{Poisson}(\lambda(t - s))$$

6 Detailed Comparison

Property	$t < s$	$t = s$	$t > s$
Distribution	$\text{Binomial}(n, \frac{t}{s})$	Deterministic	$n + \text{Poisson}(\lambda(t - s))$
Mean	$n \cdot \frac{t}{s}$	n	$n + \lambda(t - s)$
Variance	$n \cdot \frac{t}{s}(1 - \frac{t}{s})$	0	$\lambda(t - s)$
Support	$\{0, 1, \dots, n\}$	$\{n\}$	$\{n, n + 1, n + 2, \dots\}$
Depends on λ ?	NO	NO	YES

7 Examples

7.1 Example 1: Case 1 ($t < s$)

Given: $\lambda = 3$ per hour, $N_9 = 13$ (13 arrivals by 9 hours)

Find: $P(N_4 = 5 \mid N_9 = 13)$

Solution: Since $t = 4 < s = 9$, use Case 1:

$$(N_4 \mid N_9 = 13) \sim \text{Binomial}\left(13, \frac{4}{9}\right)$$

$$P(N_4 = 5 \mid N_9 = 13) = \binom{13}{5} \left(\frac{4}{9}\right)^5 \left(\frac{5}{9}\right)^8 \approx 0.203$$

This is exactly Question 4a!

7.2 Example 2: Case 3 ($t > s$)

Given: $\lambda = 2$ per hour, $N_3 = 5$ (5 arrivals by 3 hours)

Find: Distribution of N_7 (arrivals by 7 hours)

Solution: Since $t = 7 > s = 3$, use Case 3:

$$(N_7 \mid N_3 = 5) = 5 + \text{Poisson}(\lambda(t - s)) = 5 + \text{Poisson}(2 \times 4) = 5 + \text{Poisson}(8)$$

So:

- $\mathbb{E}(N_7 \mid N_3 = 5) = 5 + 8 = 13$ arrivals
- $\text{Var}(N_7 \mid N_3 = 5) = 8$
- $P(N_7 = 10 \mid N_3 = 5) = P(\text{Poisson}(8) = 5) = \frac{e^{-8} 8^5}{5!} \approx 0.092$

7.3 Example 3: Case 2 ($t = s$)

Given: $N_5 = 7$

Find: Distribution of N_5

Solution: Since $t = s = 5$, use Case 2:

$$(N_5 \mid N_5 = 7) = 7 \quad \text{with probability } 1$$

This is trivial but shows the completeness of the formula!

8 Why Case 1 and Case 3 Are Different

Key Difference

Case 1 ($t < s$): Looking at the PAST

- We're conditioning on information about the FUTURE (relative to time t)
- The n arrivals by time s are "split" between $[0, t]$ and $(t, s]$
- Answer: Binomial (NO λ needed!)

Case 3 ($t > s$): Looking at the FUTURE

- We're conditioning on information about the PAST (relative to time t)
- We have n known arrivals, plus future random arrivals
- Answer: $n + \text{Poisson}$ (DOES need λ !)

9 Common Mistakes

9.1 Mistake 1: Using the Wrong Case

Wrong: Using Binomial formula when $t > s$

Correct: Always check if $t < s$, $t = s$, or $t > s$ first!

9.2 Mistake 2: Thinking λ Always Cancels

Wrong: “The answer never depends on λ ”

Correct: When $t < s$, λ cancels. When $t > s$, you NEED λ !

9.3 Mistake 3: Wrong Support

Wrong: Thinking N_t can be any non-negative integer in all cases

Correct:

- If $t < s$: $N_t \in \{0, 1, \dots, n\}$ (can't exceed n)
- If $t > s$: $N_t \in \{n, n+1, n+2, \dots\}$ (must be at least n)

10 Why This Is Important

This question tests your understanding of:

1. **Conditional probability** for stochastic processes
2. **Independence of increments** (for Case 3)
3. **Uniform distribution of arrival times** (for Case 1)
4. **Memoryless property** of Poisson processes

It's one of the most fundamental results in Poisson process theory!

11 Summary

Key Takeaways

- Question 4b asks for a **general formula** for all t and s
- The answer has **THREE cases** depending on whether $t < s$, $t = s$, or $t > s$
- **Past** ($t < s$): $\text{Binomial}(n, t/s)$ - splits the n arrivals
- **Present** ($t = s$): Deterministic n
- **Future** ($t > s$): $n + \text{Poisson}(\lambda(t - s))$ - known + new arrivals
- Question 4a was just Case 1 with specific numbers